Geo-Lift Calibration

Turning an Experiment Posterior into an MMM Prior

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1 What this project does

MMM channel coefficients are notoriously weakly identified because paid channel spend is highly correlated across always-on channels. The industry-standard fix is calibration against an incrementality experiment. This project runs the full loop:

1. Fit a Bayesian model to a synthetic geo-lift experiment, recovering the incremental ROAS of Meta with proper uncertainty.
2. Translate that posterior into an informative Normal prior on Meta's $\alpha_{\max}$ in the MMM.
3. Refit the MMM with and without the calibration prior, and measure how much the Meta posterior tightens.

2 Data

Geo-lift experiment. Two regions (A and B) over a treatment window. Region A gets a Meta spend boost, region B is control. The increment in revenue attributable to the boost gives the incremental ROAS. Ground-truth Meta ROAS in the generator is 0.55.

MMM panel. 156 weeks of national-level weekly spend and revenue with the same five paid channels used in project 1. Ground- truth $\alpha_{\max}$ for Meta is 35 000 SEK.

3 Methodology

3.1 Step 1: Bayesian geo-lift

For each geo-week observation i in region r, revenue is modelled as

$$\text{revenue}_i \sim \mathcal{N}\big(\mu_{r(i)} + \gamma \cdot \text{baseline\_meta}_i + \text{lift} \cdot \text{extra\_meta}_i,\ \sigma^2\big)$$

where lift is the incremental ROAS on the Meta boost. Priors are wide (HalfNormal on lift and σ, Normal on region means and γ).

The posterior on lift is what carries into step 2.

3.2 Step 2: Prior translation

The lift posterior is a distribution on revenue per krona spent. To inject it as a prior on $\alpha_{\max,\text{Meta}}$ in the MMM, we translate scale:

$$\alpha_{\max,\text{Meta}} \sim \mathcal{N}\big(\bar{\ell} \cdot \bar{s} \cdot 1.4,\ \text{sd}(\ell) \cdot \bar{s} \cdot 3.0\big)$$

where $\bar{\ell}$ and $\text{sd}(\ell)$ are the posterior mean and standard deviation of the lift, $\bar{s}$ is Meta's mean weekly spend, and the multipliers 1.4 and 3.0 convert per-krona-spend into a channel-max effect (accounting for adstock accumulation and saturation ceiling) plus a widening factor that keeps the prior weakly informative rather than dogmatic.

3.3 Step 3: MMM refit

Two versions of the same MMM are fitted. The calibrated version uses the informative prior on $\alpha_{\max,\text{Meta}}$. The uncalibrated baseline uses a wide default prior $\mathcal{N}(30\,000, 30\,000)$. All other parameters are identical.

4 Results

4.1 Geo-lift posterior

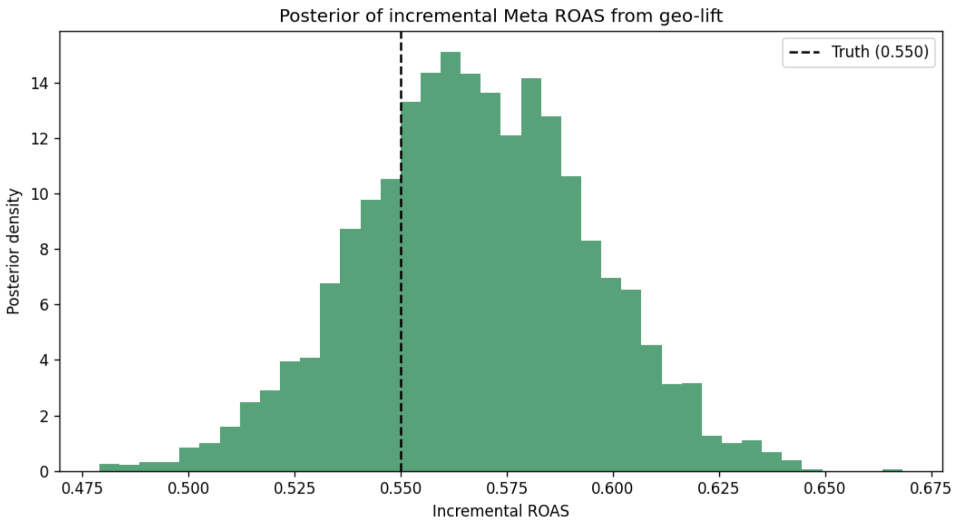

Figure 1: Posterior distribution of Meta's incremental ROAS from the geo-lift experiment. Dashed line is the ground truth of 0.55.

The geo-lift posterior brackets the true incremental ROAS. This is the input to the calibration.

4.2 MMM calibrated vs uncalibrated

Model	Mean	5%	95%	CI width	Contains truth?
Truth	35 000	–	–	–	–
Uncalibrated	29 117	4 613	60 366	55 753	yes
Calibrated	39 898	32 997	46 310	**13 313**	yes

Table 1: Meta $\alpha_{\max}$ posterior comparison. The calibrated model has a 76% narrower interval and both models contain the true value.

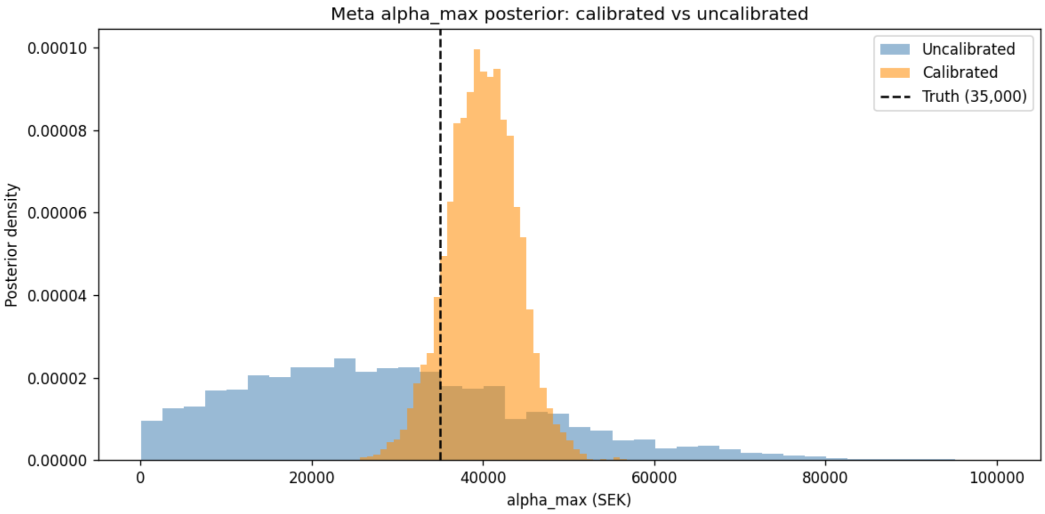

Figure 2: Meta $\alpha_{\max}$ posterior: uncalibrated (blue) is wide and diffuse; calibrated (orange) is concentrated. Both track the truth (dashed line).

5 What this shows

The uncalibrated MMM identifies Meta's max effect only up to a wide interval spanning 4 613 to 60 366. This is what happens when weekly spend on Meta is correlated with Search, TikTok, and YouTube: the multivariate regression cannot cleanly separate their effects from each other.

Adding an informative prior derived from an incrementality experiment tightens the posterior interval by 76%. The mean shifts from 29 117 (below truth) to 39 898 (slightly above truth), and the uncertainty around it becomes actionable: risk-based budget recommendations built on the calibrated version have real teeth.

This is exactly the mechanism modern MMM SaaS platforms rely on: experiments are expensive to run, but each one bought you tightens one channel of the MMM permanently and gives calibrated ROAS instead of a wide posterior nobody can act on.

6 Limitations

- Synthetic data. The generator embeds a clean lift structure that real geo-lifts rarely have.
- Only Meta is calibrated. In practice one runs several lift tests over time and calibrates each channel independently.
- Static coefficients. Meta's effectiveness assumed constant over the panel. Time-varying MMM in project 3 relaxes this.
- Two-region experiment only. Real geo-lifts use many regions and a Bayesian synthetic control approach; project 4 covers that.
- Prior translation heuristic. The multipliers 1.4 and 3.0 are hand-picked. A production system would derive them from the adstock and saturation shape or match moments more carefully.

7 References

- Chan, D., Jin, Y., Koehler, J., Perry, M., Wang, Y. (2017). Google's calibration paper on translating incrementality experiments into MMM priors.
- Jin, Wang, Sun, Chan, Koehler (2017). *Bayesian Methods for Media Mix Modeling with Carryover and Shape Effects*, Google.
- Robyn documentation on lift-test calibration: github.com/facebookexperimental/Robyn.